

A Comparative Study of Machine Learning Approaches for Intrusion Detection

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ABSTRACT – An Intrusion Detection System (IDS) is a tool that monitors network activity for signs of malicious activity. This study presents an investigation into the use of machine learning techniques for detecting exfiltration, keylogging, and OS and service scans in network activity. The study employed a supervised learning approach using labeled conversation data to identify attack IPs and using those IPS to identify attack packets in pcap files. Normal packets were also added to the training data to reduce the number of false positives. Four machine learning models (SVM, AdaBoost, decision trees, and MLP) were trained and tested on the same data for each attack type. The results showed that the models were able to accurately detect exfiltration and keylogging, but there were challenges in accurately separating OS scans and service scans. The paper also discusses the limitations and challenges of using machine learning for intrusion detection and the potential for further improvements in accuracy.

INTRODUCTION

Intrusion Detection Systems (IDS) are an important component of modern cybersecurity systems, as they provide a way to monitor the activity on a network and identify potential threats. In this research paper, we will focus on four specific types of cyber threats: Keyloggers, Exfiltration, OS scans, and Service scans.

Keyloggers are a type of malware that record every keystroke made on a computer and transmit the data to a remote location. This can allow an attacker to obtain sensitive information, such as passwords and credit card numbers, without the victim's knowledge.

Exfiltration refers to the unauthorized transfer of data from a computer or network. This can include the theft of intellectual property, trade secrets, or personal information.

OS scans and Service scans are tactics used by attackers to identify vulnerabilities in a system. OS scans involve attempting to connect to various network services and probing for weaknesses, while service scans involve identifying the specific software and versions being used on a system in order to find known vulnerabilities.

In this research paper, we will explore the role of IDS in detecting and mitigating these types of threats. We will examine the different methods that IDS systems use to identify suspicious activity, and we will discuss the challenges and limitations of these systems. By understanding the capabilities and limitations of IDS systems, we can better understand how to protect our networks and systems from these and other types of cyber threats. Our IDS was developed by testing various models to detect intrusions on a network. Support vector machine (SVM), adaptive boosting (AdaBoost), decision tree and Multilayer perceptron (MLP) were the various machine learning models used to predict the various intrusions and eventually choose the best fit one. SVMs is a versatile machine learning algorithm that finds the hyperplane in a dimensional space. Adaboost is a machine learning algorithm that utilizes an ensemble learning method that improves weak features to minimize prediction errors. Decision trees are a type of supervised machine learning algorithm that is used for classification that utilizes a labeled input to train a model. MLP is a neural network consisting of 3 layers (input, hidden and output) and uses backpropagation to train models. This paper has 5 sections. We will discuss what other authors in the field are doing in the related work's sections followed by explaining how the IDS experiment was implemented in the methodology. From there we go to the results and discussion section to put our observations into perspective. The limitations we encountered are stated in the next section and finally, in the conclusion, we summarize the results and suggest possible future avenues for research and development.

I. RELATED WORK

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patterns. After extracting features from the data using the SCAE method, they used a support vector machine (SVM) to classify the data as either good or potentially threatening. The SVM algorithm was trained on the labelled data and used to make predictions on new data. The results of the experiments indicate that the proposed SCAE+SVM method performs better at detecting intrusions compared to three other state-of-the-art methods when evaluated on the KDD Cup 99 and NSL-KDD datasets. The performance of the SCAE+SVM method on these datasets suggests that it may be a promising approach for detecting intrusions in network traffic.

[2] This study proposes a lightweight flow-based intrusion detection system (IDS) that uses statistical information about flows from SDN OpenFlow switches to analyze traffic patterns. The IDS periodically gathers this information and processes it by extracting and aggregating a set of relevant features. This allows the IDS to identify potential security threats and take appropriate actions to protect the network. They decided to use bagged trees, which is an ensemble learning method to process the statistical information and detect potential security threats.

[3] This study proposed a system that converts traffic flow characteristics into waves and uses advanced deep learning techniques based on audio/speech recognition to identify intruders. They employed a range of deep learning techniques, including long short-term memory (LSTM), deep belief networks (DBN), and convolutional neural networks (CNN). The proposed approach has been tested and found to be effective using two widely recognized and recent benchmark datasets, the NSLKDD and CIC-IDS2017 datasets. It achieved the highest detection accuracy of 84.82% on the NSLKDD dataset and 99.41% on the CIC-IDS2017 dataset.

[4] This Study uses a fused machine learning approach to detect intrusions on the network. Fused machine learning involves the use of multiple models or techniques, often in combination, to improve the accuracy and effectiveness of the final machine learning solution. The main issue with this approach is that it is more complex to implement and requires more resources. In this study, the validation accuracy of the intrusion detection system was 95.18% with a miss rate of 4.82%. In comparison, our system demonstrated an accuracy of over 99% when tested on various data sets and we also only used one machine learning model.

[5] One study used a novel system for intrusion detection using weighted class classification. The Machine learning algorithm created a classifier that was used to categorize the attacks, and the classifications were recorded in a database. A specially designed iterative algorithm was then used to improve the accuracy of the attacks.

Our approach was similar in that we both utilized machine learning to detect attacks. However, instead of employing another algorithm to enhance accuracy, we trained our model using normal network traffic to improve its performance.

[6] According to the authors, traditional single-learning models may struggle to accurately understand and classify data with complex distribution patterns, particularly when the data includes rare or unique patterns of intrusions that have a small number of samples. This can make it difficult for these models to accurately detect and classify these types of intrusions. They proposed using a big Data-based Hierarchical Deep Learning System (BDHDLs) to improve the performance of machine learning IDS. The BDHDLs approach uses both behavioral features and content features to analyze both the characteristics of network traffic and the information contained in the payload. This allows it to understand both the structure and content of the data and make more accurate predictions about potential security threats.

[7] The author suggests implementing two layers of intrusion detection systems (IDS). The first layer will categorize network connections based on the type of service being used and identify specific characteristics that will improve the ability to detect malicious activity. The second layer will utilize these characteristics and use pattern recognition techniques to determine whether the activity is normal or an attack. The experimental results demonstrate that the proposed IDS is superior to other related IDSs in detecting network intrusions. With just four features, it achieved a detection rate of 97.5%, a false alarm rate of 0.001, an MCC of 95.7%, and an overall accuracy of 99.8%.

[8] The Author presents a novel approach for detecting anomalies in network traffic using machine learning techniques. The approach involves using a distance function to determine the similarity between two feature vectors and applying a feature transformation technique to reduce the complexity of the features. The article discusses the use of various classifier algorithms on two datasets (KDD and NSL-KDD) and compares the performance of the proposed approach with other anomaly detection methods that have been previously proposed in the study and the results of the experiments suggest that the proposed approach is more effective than the other methods considered.

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...false detections in the results of the detection process. The authors propose a network intrusion detection algorithm that combines hybrid sampling with a deep hierarchical network. The algorithm begins by using one-side selection to reduce noise samples in the majority category and increase minority samples through the use of the Synthetic Minority Over-sampling Technique. This creates a balanced dataset that allows the model to learn the features of minority samples and reduces the training time. The algorithm then employs a convolution neural network to extract spatial features and a Bi-directional long short-term memory to extract temporal features, creating a deep hierarchical network model. The proposed algorithm was tested on the NSL-KDD and UNSW-NB15 datasets, with classification accuracy reaching 83.58% and 77.16%, respectively.

[10] In this study, the authors investigate the use of a deep neural network (DNN) as a flexible and effective intrusion detection system (IDS) for detecting and classifying unpredictable cyberattacks. They compare the performance of DNNs with other classical machine learning classifiers on several publicly available malware datasets. The DNN models are tested on the KDDCup 99 data set and the one that performs well on the KDDCup 99 dataset is then applied to other datasets, such as NSL-KDD, UNSW-NB15, Kyoto, WSN-DS, and CICIDS 2017. The authors find that DNNs perform better than classical machine learning classifiers in these tests. Finally, they propose a hybrid DNN framework called scale-hybrid-IDS-AlertNet that can be used in real-time to monitor network traffic and host-level events to proactively alert of possible cyberattacks.

II.

III. METHODOLOGY

A supervised learning approach for ML detection was utilized, this was accomplished by observing labeled conversation data and identifying the attack IPs, using this information the attack packets in the pcap files could then be identified. For each attack type, three associated PCAP files containing the respective attacks were used in the training process. In each case, the pcap file with the greatest number of attack packets was chosen as the training reference.

When this experiment was originally conducted the ML models had a lot of false positives, to alleviate this, normal packets were also added to the training data. Wireshark was set to capture packets while normal activity was conducted e.g. youtube, web browsing, web searches, etc. (See Fig 1.). Using this method, approximately Eighty thousand (80000) packets were then saved to a pcap file and exported as a CSV file using the same columns as the extracted CSV file from the attack pcaps. All packets in this CSV file were labeled as normal packets by setting the attack column as 0. This data was added to each extracted attack dataset and the models were retrained. By doing this the number of false positives dropped dramatically.

There are 4 ML models used, SVM, AdaBoost, decision trees and MLP. Each model was trained on the same data for each attack type and tested on the same data to ensure a fair comparison. They were then scored against each of the three (3) pcap files for each of the 4 types of attacks, and the results were collected.

For some of the columns e.g bytes since the last psh flag, their values can get really large compared to other fields, therefore scaling was applied using the standard scaler. This reduced the wide range of the data to prevent one column from overpowering the others due to much larger values.

The pcap file with the largest number of attack packets for each attack type was exported as a CSV, the resultant CSV file is split percentage-wise Train/Test 75%/25%. After training on the first 75% it was scored against the next 25%, if a good test accuracy is observed, the models are then tested against the other 2 extracted labeled CSV files, if good test accuracies are again observed, the ML model along with the scalar model is exported for later use as ".pickle" files. These can then be imported to be used in the prediction script.

Exfiltration, these types of attacks include the transfer of data from the victim's computer, therefore features that indicate large data transfer at the packet level were mainly considered (Bytes sent since last psh flag, bytes in flight, Length) e.g. After the analysis was found that bytes in flight tended to remain in the 1200 to 1500 range for normal packets but vary a lot or be much higher than that for attack packets. The final list of features chosen for this type of attack is source port, destination port, protocol, Length, window, TCP flag, bytes in flight, bytes sent since last psh flag, time since the previous frame in this TCP stream. (10 features).

Keylogging packets would generally be smaller in length, and would likely send a set amount of data periodically, which would mean a particular port sending a low packet length with the same durations between packets could be an indication of keylogging. The final list of features chosen for this type of attack is source port, destination port, protocol, length, window, TCP, flag, bytes in flight and time since the previous frame in the stream.(9 features).

OS and Service scan packets would be packets that are ^[a] sent to get a response that reveals information about the system. The attacker would rely on information such as TTL, TCP flags, ports etc. to determine what OS or service is running. The final list of features for this kind of attack is source port, destination port, protocol, length, window, TCP flags, bytes in flight, time since the last frame(TCP), time since the last frame(UDP). (9 features)

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with the same packet configurations.

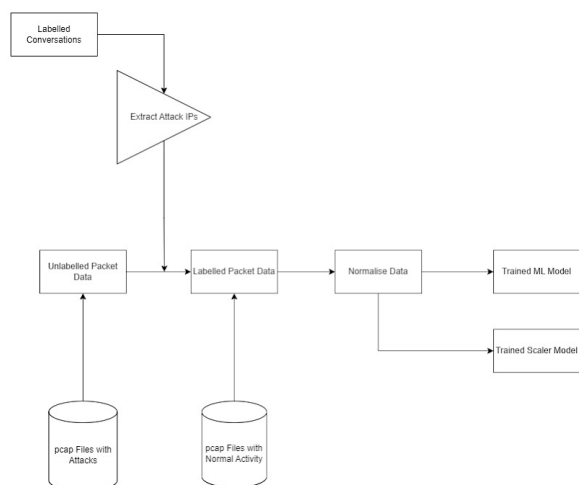


Fig 1. IDS ML model Training Procedure

The IDS detection script allows the user to perform two types of checks, a continuous check and an instance check. The continuous check entails a constant scanning of the user's chosen interface, e.g "eth0" will be captured and sent to the intrusion detection system. With instance check mode you input the file path of the pcap file with suspected attacks and specify which packet by using its index, the IDS script will then import that packet and run it through the prediction models and output a prediction in the console, it will also append the full packet info to the log file if it detects an attack.

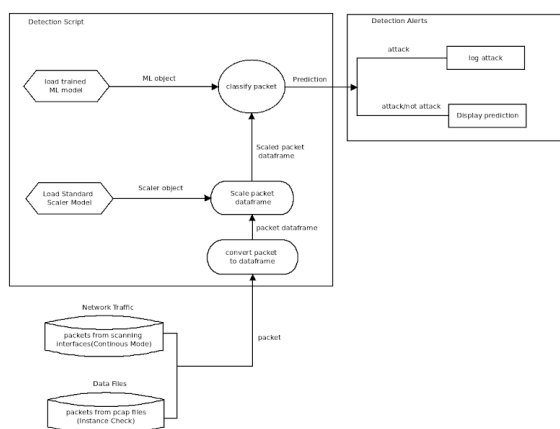


Fig 2. How The detection mechanism works

IV. RESULTS AND DISCUSSION

The results were obtained by training each model against each type of attack, the resultant model was then tested against the entirety of the other 2 pcap files for accuracy. We can observe that even after training with only one pcap file, the accuracy against the other two separate pcap files is always above 98% for all models. Adaboost performed the best consistently over all types of attack with accuracy above 99.5% there is even a case where the accuracy was 100% for exfiltration.

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SVM	Keylogging	99.89	99.86	99.96
Adaboost	Keylogging	99.97	99.97	99.98
MLP(10,7,4)	Keylogging	99.89	99.85	99.93
Decision Tree	Keylogging	99.99	99.98	99.99
SVM	OS Scan	99.58	99.61	99.62
Adaboost	OS Scan	99.7	99.62	99.62
MLP(10,7,4)	OS Scan	99.91	99.9	99.9
Decision Tree	OS Scan	99.99	99.98	99.98
SVM	Service Scan	99.66	99.57	99.6
Adaboost	Service Scan	99.92	99.92	99.92
MLP(10,7,4)	Service Scan	99.86	99.64	99.51
Decision Tree	Service Scan	99.99	99.24	98.56

Key	Number
PCAP Files	
IoT Dataset data theft 00001 20180618110503	1
IoT Dataset data theft 00002 20180618111101	2
IoT Dataset data theft 00003 20180618111233	3
IoT Keylogging 00001 20180619135903	1
IoT Keylogging 00002 20180619140719	2
IoT Keylogging 00003 20180619141524	3
IoT Dataset OSScan 00001 20180521140502	1
IoT Dataset OSScan 00002 20180521143228	2
IoT Dataset OSScan 00003 20180521150020	3
IoT Dataset ServiceScan 00001 20180515102711	1
IoT Dataset ServiceScan 00001 20180521150502	2
IoT Dataset ServiceScan 00001 20180521200502	3

Fig. 3. Results showing the accuracy after training each model against each attack type

Although the accuracy in the training and testing was 98% when actually running the models in a physical network environment a lot of false positives are observed, this was alleviated originally by adding in the normal packets in the dataset and retraining. However, after implementing the Man in the Middle attack using a windows OS, false positives were again observed, it is theorized that these can be mitigated by collecting packets during the man in the middle configuration with the Windows OS and adding this into the training dataset.

One previous study that used a similar procedure to the one described in this paper obtained an accuracy rate of 97.5%. While this is still a high accuracy rate, it is not as high as the accuracy rate achieved in this project (over 99%). Because of the limited nature of the dataset used in our research, it is hard to compare however with such a high accuracy this supports the idea that machine learning techniques are very effective in identifying threats at least within the confines of environments that mirror the training datasets. Therefore the question of how well an ML model performs in a novel network environment is really a question of how well the training datasets are.

In related research on this topic, many of them have used machine learning approaches for anomaly detection. Some of these studies used multiple machine learning models to improve accuracy. In contrast, others employed deep learning techniques by transforming packets into waves and using these to classify the data as either normal or an attack. A notable difference between these approaches and the one proposed in this paper is that previous studies had access to well-known attack datasets for training their models.

V. LIMITATION

One major limitation is the lack of attributes at the packet level to accurately classify packets, Deep packet inspection was the most accurate method to detect threats however the use of packet-level encryption to protect private information, has severely limited the classification capabilities at the packet level. This is why flows tend to be used as more information can be garnered from a group of related packets vs just one packet by itself.

Initially, we trained the model using a limited data set of pcap files containing examples of keylogging, OS scanning, service scanning, and data exfiltration attacks. While the model was able to learn to detect these attacks, it also generated a large number of false positives when presented with normal network activity. To address this issue, we decided to augment the training dataset with additional examples of normal network activity that we collected ourselves. This provided the model with a reference point for distinguishing between normal and malicious activity and helped reduce false positives. However when used in a man-in-the-middle setup where a separate OS was sending packets to it, false positives were again observed, this can be alleviated by repeating this step, collecting packets during the man-in-the-middle session and marking them as normal traffic to reduce false positives.

If we had more time, we would have preferred to use a larger and more diverse training dataset that included more examples of various types of attacks. This would have allowed the model to learn how to recognise and detect a wider range of attacks. However, due to limitations on our time, we were unable to obtain additional pcap data for these attacks.

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VI. CONCLUSION

In conclusion, this research paper investigated the use of machine learning in intrusion detection at the packet level, focusing on four types of cyber threats: keyloggers, exfiltration, OS scans, and service scans. The results showed that machine learning-based IDS systems could accurately detect keyloggers and exfiltration in both training and test data. However, there were challenges in accurately detecting and distinguishing between OS scans and service scans. Further research is needed to improve the accuracy of machine learning-based IDS systems in detecting these types of threats.

Further improvements could involve using an ensemble approach to reduce false positives, having the script automatically train in any network environment it is placed in to enable more accurate detections and using much larger and more varied datasets that encompass typical network environments.

Despite the limitations, this study demonstrates the potential for machine learning to enhance the capabilities of IDS systems and highlights the importance of ongoing research in this field.

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[a] forgot to use ttl as feature in OS and service scan, too late to add it?